

Unnamed_Unit

Unit 2 - Bringing Ideas to Life

PDF Documents

PDF Document 1: 1738064627520-Chapter 7 - unit 2 - Bringing Ideas to Life.pdf

This PDF document is included in the unit materials.



UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

STORY TELLING - DIGITAL ROCK-PAPER-SCISSORS GAME

In Dawn's fifth-grade class, everyone loved the weekly Rock-Paper-Scissors tournament. But some students felt shy playing in front of others.



Seeing this, Dawn introduced a project using the MODI Master kit. "We're going to make a digital Rock-Paper-Scissors game," she explained. "You'll use the Dial module to secretly choose rock, paper, or scissors and the Button module to confirm your choice on a screen."

The students were excited to try it out. With the new digital game, even the shyest could join in without feeling nervous. By the end of the class, everyone was playing and having fun together. The digital game helped everyone feel included and showed how technology can make games even more enjoyable.

DID YOU KNOW?

The digital rock-paper-scissors game is not a new concept. There are arcade centers where digital rock-paper-scissors machines are installed, allowing people to remotely play the game with friends.



UNDERSTANDING DIAL MODULE

Now, let's apply this understanding to our game setup as below.

1% TO 33%

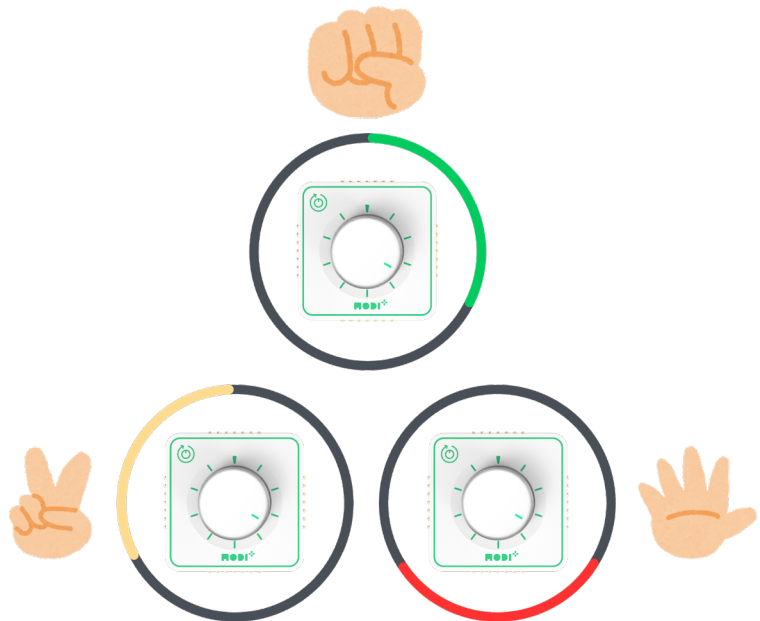
- Turning the dial within this range selects Rock.

34% TO 66%

- Positioning the dial here chooses Paper.

67% TO 100%

- Rotating the dial further within this range picks Scissors.



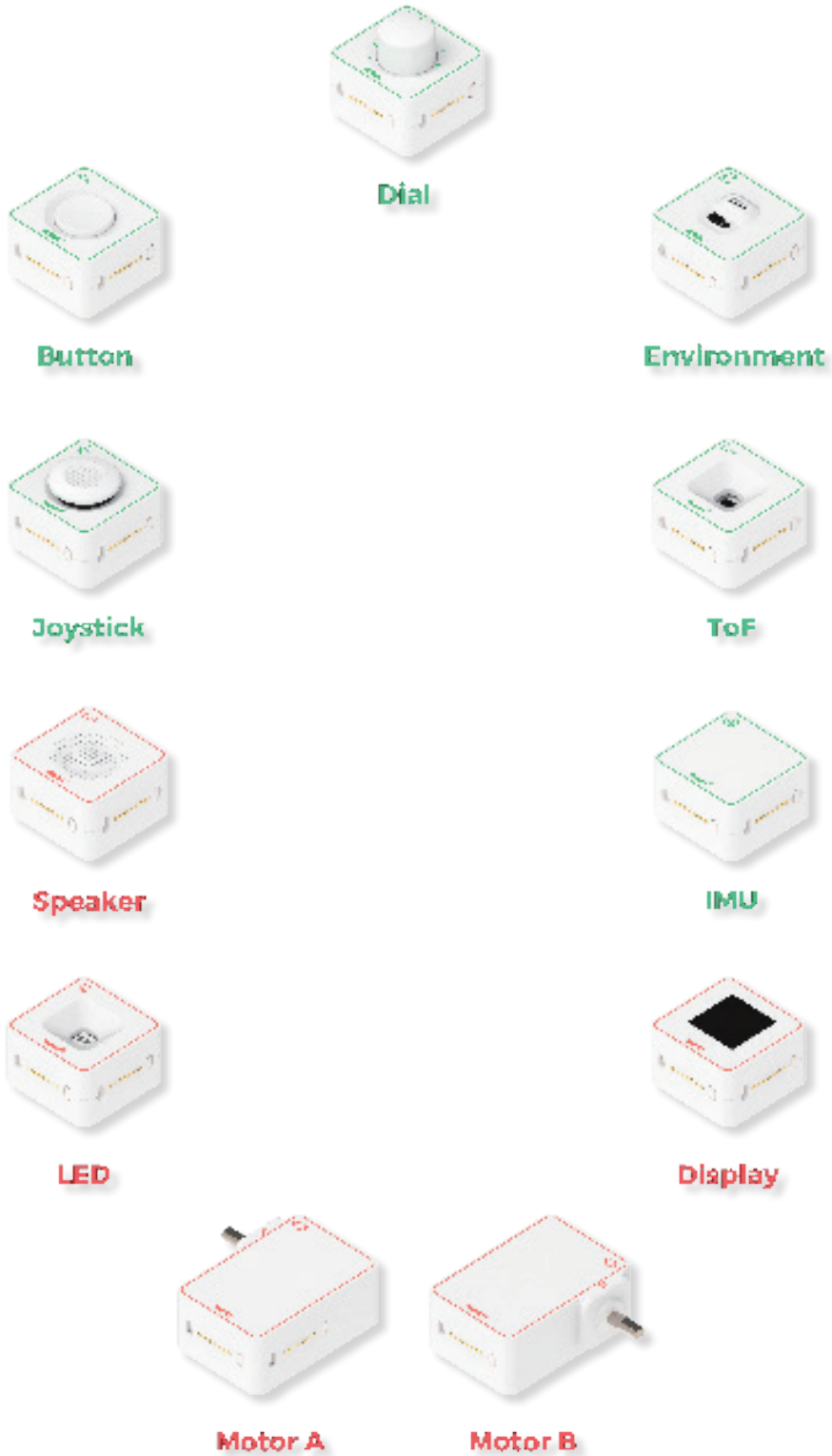
This setup allows players to easily select their moves by simply turning the dial. The precise detection of the dial's position ensures each choice is accurately registered.



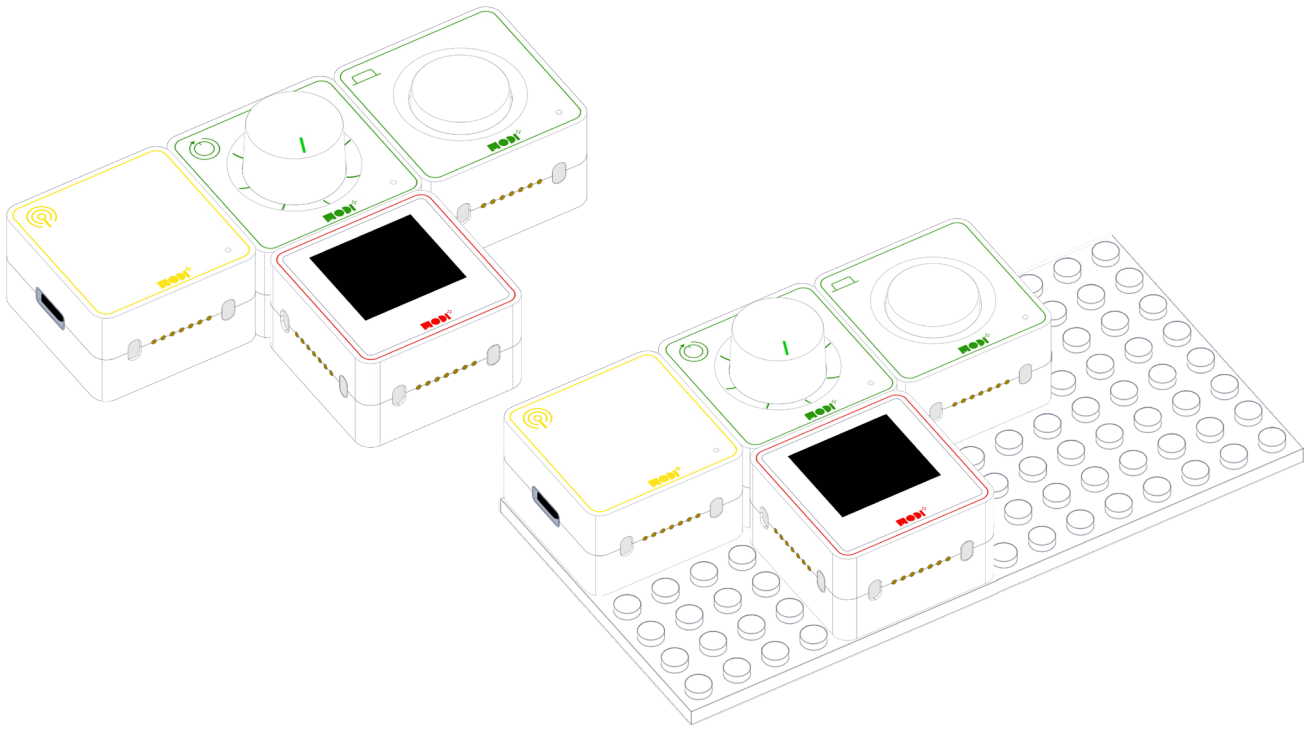


Required Modules & Brainstorming

Based on the story, which MODI modules do you need for this creation?



HOW DOES THE CREATION LOOK LIKE?

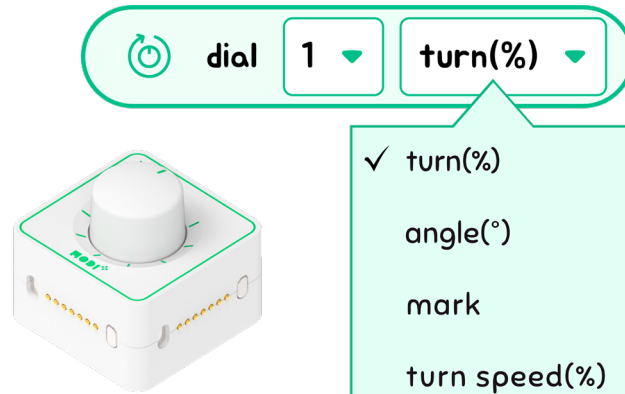


Buttons and dials are input devices that we frequently use in our daily lives.



Module Functionality

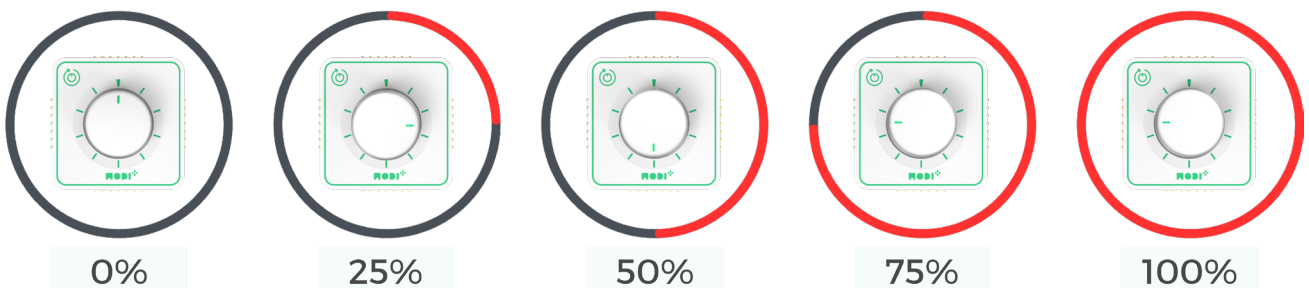
Let's look detail about the Dial module.



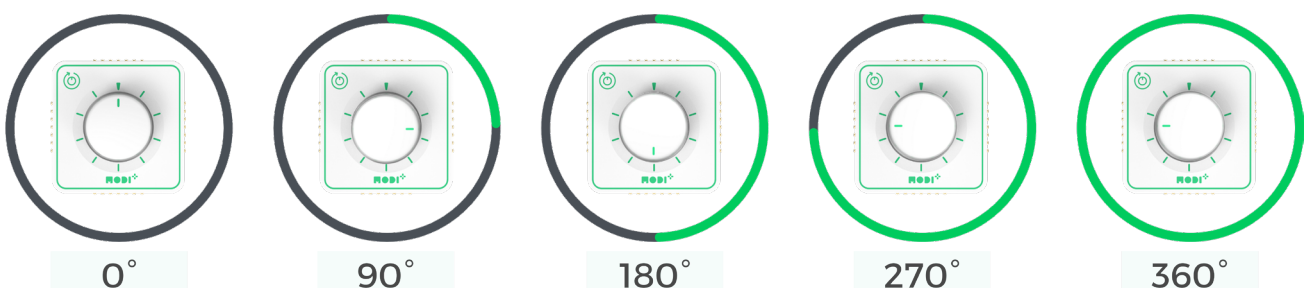
FUNCTIONS IN DETAIL

1. Turn (%) - Range: 0% ~ 100% This just means moving the dial around, either to the right or left.	2. Angle (°) - Range: 0~360° This is how far you've moved the dial from the default (0°), measured in degrees.
3. Mark - Range: 1~10 Marks on the dial show where it can be turned to, like the numbers on a clock.	4. Turn Speed (%) - Range: 0% ~ 100% This tells us how fast you are turning the dial, in percentage.

TURN (%)



ANGLE (°)



The Dial module can rotate from 0 to 360 degrees, similar to turning a knob full circle. For simplicity, we'll consider a full rotation as 100%. Therefore, a half-turn, or 180 degrees, represents 50% of a full rotation.